

Multiple Choice (10 marks)

1. The Large Magellanic Cloud is ...
 - (a) a dwarf galaxy orbiting the Milky Way.
 - (b) the closest planetary nebula to the Earth.
 - (c) a bright star cluster discovered by Magellan.
 - (d) the outer arm of the Milky Way named after Magellan.
2. Which factor is most important in determining the history of volcanism and tectonism on a planet?
 - (a) size of the planet
 - (b) presence of an atmosphere
 - (c) distance from the sun
 - (d) rotation period
3. Which of the following most likely explains why Venus does not have a strong magnetic field?
 - (a) Its rotation is too slow.
 - (b) It has too thick an atmosphere.
 - (c) It is too large.
 - (d) It does not have a metallic core.
4. The Pleiades is an open star cluster that plays a role in many ancient stories and is well-known for containing ... bright stars.
 - (a) 5
 - (b) 7
 - (c) 9
 - (d) 12
5. Why is the sky blue?
 - (a) Because the molecules that compose the Earth's atmosphere have a blue-ish color.
 - (b) Because the sky reflects the color of the Earth's oceans.
 - (c) Because the atmosphere preferentially scatters short wavelengths.
 - (d) Because the Earth's atmosphere preferentially absorbs all other colors.

True / False (10 marks)

Answer True or False

6. True or False: Hydrogen is the most abundant element found in the atmospheres of the Jovian planets.
7. True or False: Mars experiences more extreme seasons due to its more eccentric orbit compared to Earth's nearly circular orbit.
8. True or False: The atmosphere on Mars is too thin to effectively trap a significant amount of heat, limiting the greenhouse effect.
9. True or False: Moons contribute to the stability of particles within rings, create gravitational effects at ring edges, and form gaps between rings.
10. True or False: When the angular separation of two stars is smaller than the angular resolution of the eye, they will be distinctly visible as two separate stars.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the phenomena associated with orbital resonance in astronomy, and analyze why the breaking of small Jovian moons to form ring materials is not a consequence of this process.
12. Discuss the concept of a black body in astronomy, including its definition, significance in radiation calculations, and implications for understanding stellar properties.

End of Paper